25900267 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBOTA08 - SOIL SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Questions

Answer any **10** questions

Each question carries **2** marks

1. How does soil texture influence pore space? [U] / [CO1, CO2, CO3, CO4, CO5]
2. Name one leguminous crop used as green manure. [U] / [CO1, CO2, CO3, CO4, CO5]
3. Which of the following is considered a macronutrient for plants? [U] / [CO1, CO2, CO3, CO4, CO5]
4. Which form of soil water drains quickly after rainfall? [U] / [CO1, CO2, CO3, CO4, CO5]
5. Give one example of a nitrogenous fertilizer. [U] / [CO1, CO2, CO3, CO4, CO5]
6. Define soil fertility. [U] / [CO1, CO2, CO3, CO4, CO5]
7. Name one characteristic of good soil tilth. [U] / [CO1, CO2, CO3, CO4, CO5]
8. Short note on Jenny's conceptof soil formation. [U] / [CO1, CO2, CO3, CO4, CO5]

9.	What are primary minerals? Give one example.	[U] / [CO1, CO2, CO3, CO4, CO5]
10.	What is soil science?	[U] / [CO1, CO2, CO3, CO4, CO5]
11.	What is the molecular structure of a water molecule?	[U] / [CO1, CO2, CO3, CO4, CO5]
12.	Explain soil texture.	[U] / [CO1, CO2, CO3, CO4, CO5]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **5** marks

13.	Differentiate soil profile and soil horizon.	[U] / [CO1, CO2, CO3, CO4, CO5]
14.	How does water behave in soil?	[U] / [CO1, CO2, CO3, CO4, CO5]
15.	What is soil air, and how does its composition differ from atmospheric air?	[U] / [CO1, CO2, CO3, CO4, CO5]
16.	What does the term "time" refer to in soil formation, and why is it classified as a passive factor?	[U] / [CO1, CO2, CO3, CO4, CO5]
17.	What is the definition of weathering in geology?	[U] / [CO1, CO2, CO3, CO4, CO5]
18.	What factors influence the particle density of a soil?	[U] / [CO1, CO2, CO3, CO4, CO5]
19.	Mention any two soil factors affecting available water.	[U] / [CO1, CO2, CO3, CO4, CO5]
20.	Name and briefly describe four common soil structure types.	[U] / [CO1, CO2, CO3, CO4, CO5]
21.	What are the differences between Monomineralanic and polymineralanic rocks.	[U] / [CO1, CO2, CO3, CO4, CO5]
22.	Discuss how soil temperature influences overall crop growth	[U] / [CO1, CO2, CO3, CO4, CO5]

[5x8 = 40]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

23.	Discuss the overall significance of thermal properties of soil in agriculture.	[U] / [CO1, CO2, CO3, CO4, CO5]
24.	What is enriched compost? Discuss how compost can be enriched with essential nutrients like nitrogen, phosphorus, potassium, and bioinoculants.	[An] / [CO1, CO2, CO3, CO4, CO5]

[10x1 = 10]